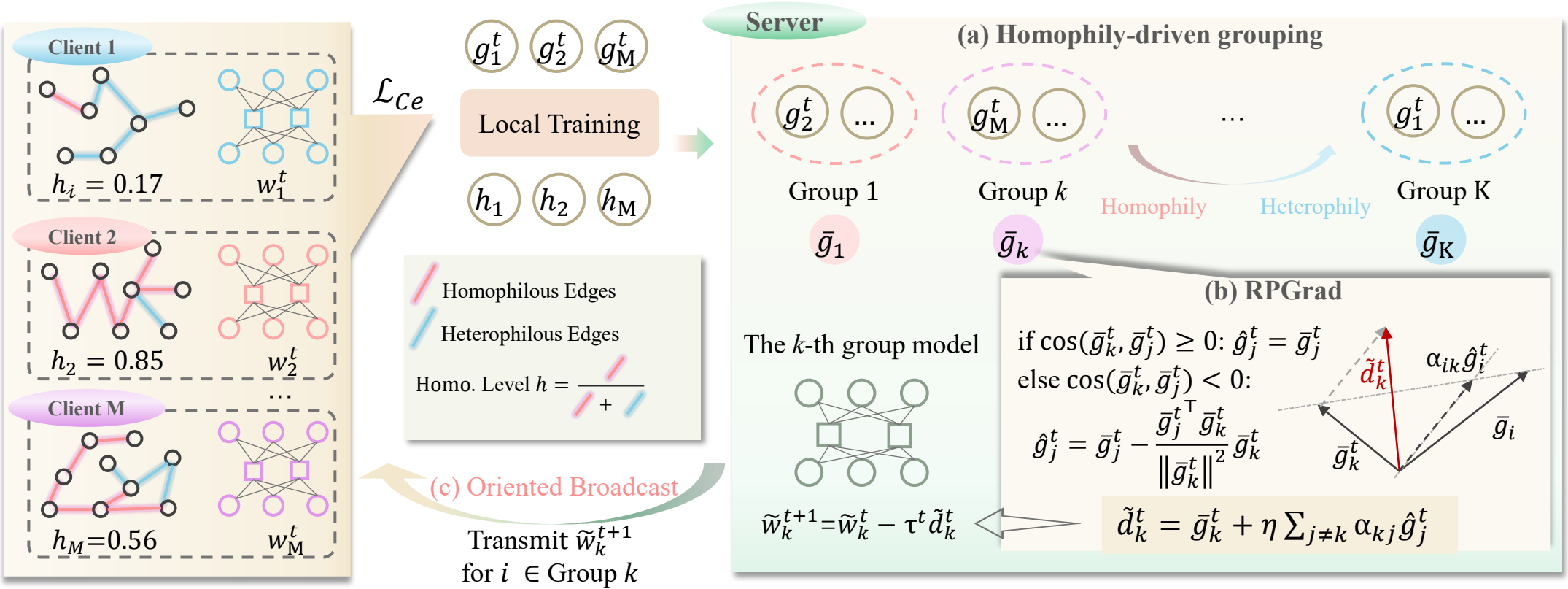
Client 1
$h_i = 0.17$
w_1^t
Client 2
$h_2 = 0.85$
w_2^t
Client M
$h_M = 0.56$
w_M^t
$\mathcal{L}_{Ce}$
g_1^t g_2^t g_M^t
Local Training
h_1 h_2 h_M
Homophilous Edges
Heterophilous Edges
Homo. Level h =
Server
(a) Homophily-driven grouping
Group 1
Group k
Group K
Homophily
Heterophily
$\bar{g}_1$
$\bar{g}_k$
$\bar{g}_K$
The k-th group model
(b) RPGrad
if $\cos(\bar{g}_k^t, \bar{g}_j^t) \geq 0$: $\hat{g}_j^t = \bar{g}_j^t$
else $\cos(\bar{g}_k^t, \bar{g}_j^t) < 0$:
$\hat{g}_j^t = \bar{g}_j^t - \frac{\bar{g}_j^{t\top} \bar{g}_k^t}{\|\bar{g}_k^t\|^2} \bar{g}_k^t$
$\tilde{d}_k^t$
$\alpha_{ik} \hat{g}_i^t$
$\bar{g}_i$
$\bar{g}_k^t$
$\tilde{d}_k^t = \bar{g}_k^t + \eta \sum_{j \neq k} \alpha_{kj} \hat{g}_j^t$
$\tilde{w}_k^{t+1} = \tilde{w}_k^t - \tau^t \tilde{d}_k^t$
(c) Oriented Broadcast
Transmit $\tilde{w}_k^{t+1}$
for $i \in$ Group k